

Week 1

What is a Database?

Definition: A database is a structured collection of data that can be accessed, managed, and updated. It serves as a repository for organized data that supports efficient retrieval and management.

Examples:

A customer directory in an e-commerce application.

A hospital's patient records system.

An educational institution's student database.

What is a Database?

Characteristics of Databases:

1

- Organized and structured.

2

- Persistent and stored electronically.

3

- Accessible by multiple users or applications.

What is a Database Management System (DBMS)?

❑ **Definition:** A Database Management System (DBMS) is software designed to define, manipulate, retrieve, and manage data in a database.

❑ **Core Functions of DBMS:**

- ❑ **Data Storage:** Efficiently stores large amounts of data.
- ❑ **Data Retrieval:** Allows quick access to data using queries.
- ❑ **Data Manipulation:** Supports operations like insertion, deletion, and updating of records.
- ❑ **Data Security:** Ensures authorized access and protects data integrity.

Popular DBMS Examples:

- Relational DBMS: MySQL, PostgreSQL, Oracle Database.
- NoSQL DBMS: MongoDB, Cassandra, DynamoDB.
- Cloud DBMS: Google Cloud Spanner, Amazon RDS.

Key Differences Between Data and Databases

Aspect	Data	Database
Definition	Raw, unprocessed facts and figures.	Organized collection of structured data.
Example	“John, 25”	A table with columns: Name, Age.
Purpose	Basic unit of information.	Stores and manages related data.

Components of a DBMS

- **Database Engine:** Core software component that performs storage and retrieval tasks.
- **Query Processor:** Interprets user queries (e.g., SQL commands) and converts them into actionable tasks.
- **Data Definition Language (DDL):** Defines database schema and structure (e.g., CREATE, ALTER).
- **Data Manipulation Language (DML):** Performs data operations like insertion, deletion, and updates (e.g., INSERT, UPDATE, DELETE).
- **Transaction Manager:** Ensures data integrity and consistency during concurrent user access.

Types of DBMS

- **Relational DBMS (RDBMS):**
 - Data is organized in tables with rows and columns.
 - Follows a strict schema and supports SQL.
 - Example: MySQL, PostgreSQL.
- **NoSQL DBMS:**
 - Designed for unstructured or semi-structured data.
 - Uses document, key-value, or graph data models.
 - Example: MongoDB, Redis.

Types of DBMS(Cont.)

- **Hierarchical DBMS:**
 - Data is organized in a tree-like structure.
 - Example: IBM's IMS.
- **Network DBMS:**
 - Data is organized in a graph, with multiple parent-child relationships.
 - Example: Integrated Data Store (IDS).
- **Object-Oriented DBMS (OODBMS):**
 - Supports object-oriented data models.
 - Example: ObjectDB.

ADVANTAGES OF USING DBMS

◉ Data Centralization:

- Provides a single location for data storage, reducing redundancy.

◉ Data Integrity and Consistency:

- Enforces rules to ensure data remains accurate and consistent.

◉ Data Security:

- Implements user authentication and access controls.

ADVANTAGES OF USING DBMS

◎ **Backup and Recovery:**

- Ensures data is protected and can be restored in case of failure.

◎ **Concurrent Access:**

- Supports multiple users accessing the database simultaneously without conflicts.

Real-World Applications of Databases

- **Banking:**
 - Customer accounts, transactions, and loan processing.
- **Healthcare:**
 - Patient records, prescriptions, and billing.
- **Retail:**
 - Inventory management, customer orders, and sales analytics.
- **Education:**
 - Student management, course registrations, and grading systems.
- **Social Media:**
 - User profiles, posts, and interactions.

Evolution of Database Systems

- **Early Systems:**
 - Flat-file systems and manual data storage techniques.
- **Hierarchical and Network Models:**
 - Emerged in the 1960s to handle more complex data relationships.
- **Relational Databases:**
 - Introduced by E.F. Codd in the 1970s; became the standard for structured data.
- **NoSQL and Cloud Databases:**
 - Developed in the 21st century to handle large-scale, unstructured data and distributed systems.